Project Objective

To calculate the interest rate sensitivity (DV01) of a long position in a Mortgage-Backed Security (MBS) TBA contract and to structure a duration-neutral macro hedge using a combination of interest rate swaps to fully offset this risk.

Background & Data

The portfolio manager holds a long MBS position and needs to hedge against rising interest rates using the following transaction details and market parameters:

- **TBA Position Details:**
  - Direction: Long
  - Face Value: $100 million
  - Security: 10-Year Fannie Mae 5.5% TBA
  - Market Price: 99.34375
  - Settlement Date: August 13, 2026
  - Prepayment Assumption: 6% Constant Prepayment Rate (CPR)
- **Hedging Instruments & Mandate:**
  - Instruments: 5-Year and 10-Year Pay-Fixed SOFR Swaps
  - 5-Year Swap DV01: 0.0455% of notional per 1 bp
  - 10-Year Swap DV01: 0.0825% of notional per 1 bp
  - Risk Allocation: Exactly 40% of the total hedge DV01 must be sourced from the 5-Year Swap, and 60% from the 10-Year Swap.

Deliverable

A final calculation providing two distinct outputs:

1. The exact value the TBA portfolio loses when interest rates increase by one basis point (the TBA DV01).
2. The exact Notional Amounts of the 5-Year and 10-Year Pay-Fixed SOFR Swaps required to execute the mandated duration-neutral perfect hedge.